Set-up

X is the daily log return, $V = \$1{,}000{,}000$, and the linearized loss is $L = -VX$. $\mathrm{VaR}_\alpha(L) = \inf\{x : F_L(x) \geq \alpha\}$, where F_L is the cdf of L. Sample: $N = 2555$ returns, losses $l_1, \ldots, l_N$, order statistics $l_{(1)} \leq \ldots \leq l_{(N)}$.

(1) Empirical VaR

The empirical cdf is $F_N(x) = \frac{1}{N}\#\{i : l_i \leq x\}$. It is a step function with $F_N(l_{(k)}) \geq k/N$, and $F_N(x) < k/N$ for $x < l_{(k)}$. So the smallest x with $F_N(x) \geq \alpha$ is $l_{(k)}$ with k the smallest integer such that $k/N \geq \alpha$:

$$\boxed{\mathrm{VaR}_\alpha = l_{(\lceil N\alpha \rceil)}.}$$

At $\alpha = 0.9999$: $\lceil 2555 \times 0.9999 \rceil = \lceil 2554.74 \rceil = 2555$, the largest loss. The same holds for every $\alpha > 2554/2555 = 0.99961$.

(2) Normal with EWMA moments

Given $X \sim N(\mu, \sigma^2)$, $L = -VX \sim N(-V\mu, V^2\sigma^2)$, so $F_L(x) = \Phi((x + V\mu)/(V\sigma))$. This is continuous and strictly increasing, so

VaR_α at solves $F_L(x) = \alpha$:

$$\boxed{VaR_\alpha = -V\mu + V\sigma\Phi^{-1}(\alpha).}$$

At $\alpha = 0.9999$, $\Phi^{-1}(0.9999) = 3.719016$: VaR $= -1{,}210.10 +$ $7{,}227.07 \times 3.719016 = 25{,}667.50$.

(3) GEV, block maxima

Scaling.

The fit is done on the block maxima of the losses in \$ thousand, M_n/k with $k = 1000$, giving $M_n/k \sim H_{\xi, \mu_k, \sigma_k}$. Then

$$P[M_n \le x] = H_\xi\left(\frac{x/k - \mu_k}{\sigma_k}\right) = H_\xi\left(\frac{x - k\mu_k}{k\sigma_k}\right) = H_{\xi, k\mu_k, k\sigma_k}(x),$$

so the dollar parameters are $\mu = k\mu_k$, $\sigma = k\sigma_k$, with the same ξ.

From the maximum to one day.

For i.i.d. losses, $P[\max\{L_1, \dots, L_n\} \le x] = F_L(x)^n$, so with the GEV fit

$$F_L(x) = H_{\xi, \mu, \sigma}(x)^{1/n}.$$

Solving for VaR.

Set $F_L(x) = \alpha$, i.e. $H_{\xi,\mu,\sigma}(x) = \alpha^n$, and write $z = (x-\mu)/\sigma$:

$$\exp\left(-(1+\xi z)^{-1/\xi}\right) = \alpha^n \Rightarrow$$

$$(1+\xi z)^{-1/\xi} = -n\log\alpha \Rightarrow 1+\xi z = (-n\log\alpha)^{-\xi},$$

$$\boxed{\mathrm{VaR}_\alpha = \mu + \frac{\sigma}{\xi}\left((-n\log\alpha)^{-\xi} - 1\right) = \mu - \frac{\sigma}{\xi}\left(1-(-n\log\alpha)^{-\xi}\right).}$$

$-n\log\alpha > 0$ for $\alpha < 1$, so this is defined for any $\xi \neq 0$.

scipy's sign.

genextreme has cdf $\exp\left(-(1-cz)^{1/c}\right)$, which is H_ξ with $c = -\xi$.

At $\alpha = 0.9999$.

- $n = 73$: $-73\log 0.9999 = 0.0073004$ and $0.0073004^{-0.1815} = 2.4423$, so VaR $= 20{,}958.16 + \frac{8{,}732.71}{0.1815}(1.4423) = 90{,}354$. (With unrounded parameters: 90,352.51.)
- $n = 35$: $-35\log 0.9999 = 0.0035002$ and $0.0035002^{-0.2141} = 3.3559$, so VaR $= 15{,}298.85 + \frac{8{,}246.20}{0.2141}(2.3559) = 106{,}038$. (With unrounded parameters: 106,042.15.)

(4) GP excess losses

Tail of F_L.

For $x > u$,

$$F_L(x) = P[L \le u] + P[u < L \le x]$$

$$= F(u) + (1 - F(u)) P[L - u \le x - u \mid L > u]$$

$$= F(u) + (1 - F(u)) G_{\xi,\beta}(x - u).$$

Solving for VaR.

For $\alpha > F(u)$, set $F_L(x) = \alpha$:

$$1 - G_{\xi,\beta}(x - u) = \left(1 + \frac{\xi(x-u)}{\beta}\right)^{-1/\xi} = \frac{1-\alpha}{1-F(u)}$$

$$\Rightarrow \boxed{VaR_\alpha = u + \frac{\beta}{\xi}\left(\left(\frac{1-F(u)}{1-\alpha}\right)^{\xi} - 1\right).}$$

Scaling.

The fit is on the excesses in \$ thousand. $G_{\xi,\beta_k}(y/k) = 1 - (1 + \xi y/(k\beta_k))^{-1/\xi} = G_{\xi,k\beta_k}(y)$, so $\beta = k\beta_k$ with the same ξ. genpareto's shape c is ξ itself.

At $\alpha = 0.9999$.

$u = 15{,}252.86$ (the $\lceil 2555 \times 0.95 \rceil = 2428$th order statistic) and $F(u) = 0.95$. Then $(1 - 0.95)/(1 - 0.9999) = 500$ and $500^{0.2656} = 5.2101$, so

$$\text{VaR} = 15{,}252.86 + \frac{1{,}594.30}{0.2656}(4.2101) = 135{,}632.$$

(With unrounded $\xi = 0.26555$: 135,610.18.)